

RSIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool: Concept Mapping (Medium)

Assessment Tool Weightage: 20%

Code of Conduct:

I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

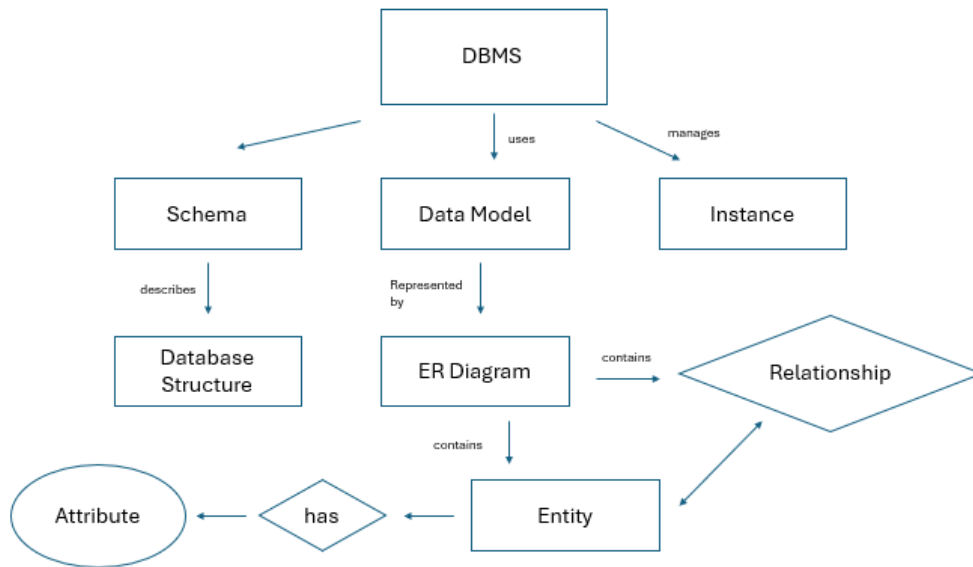
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

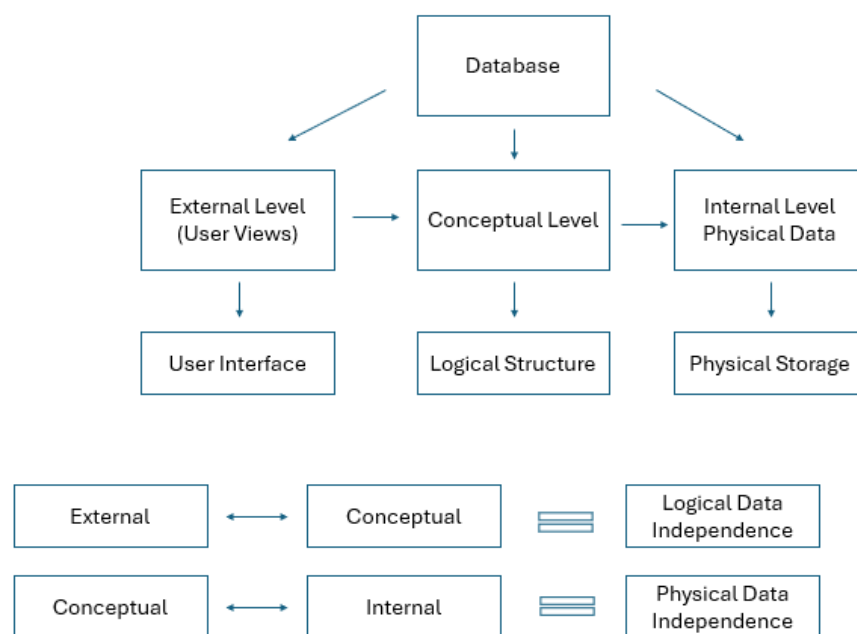
Judy Allen J

Q1. Construct a concept map linking the following terms:

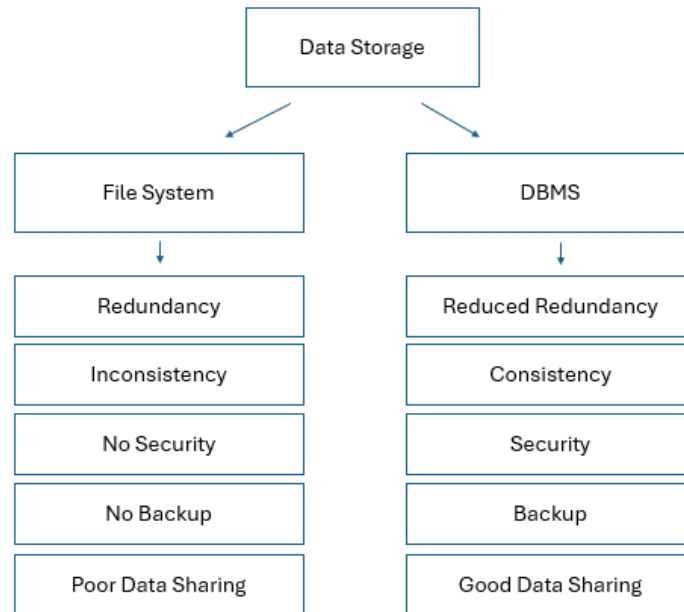
DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship



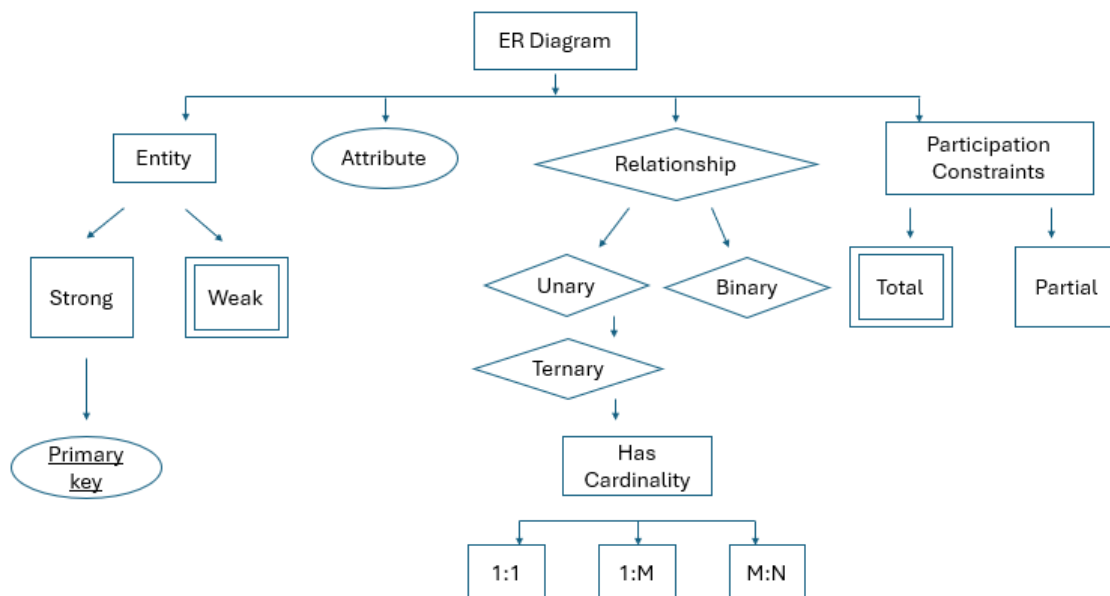
Q2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.

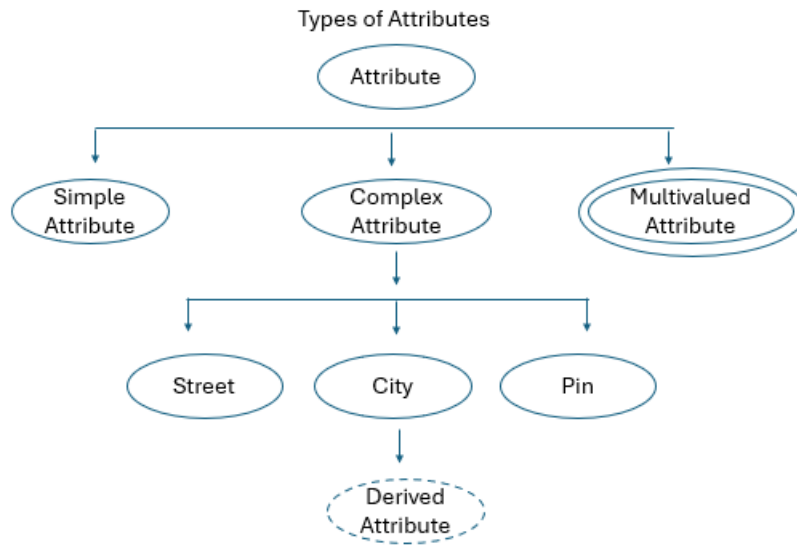


Q3. Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.

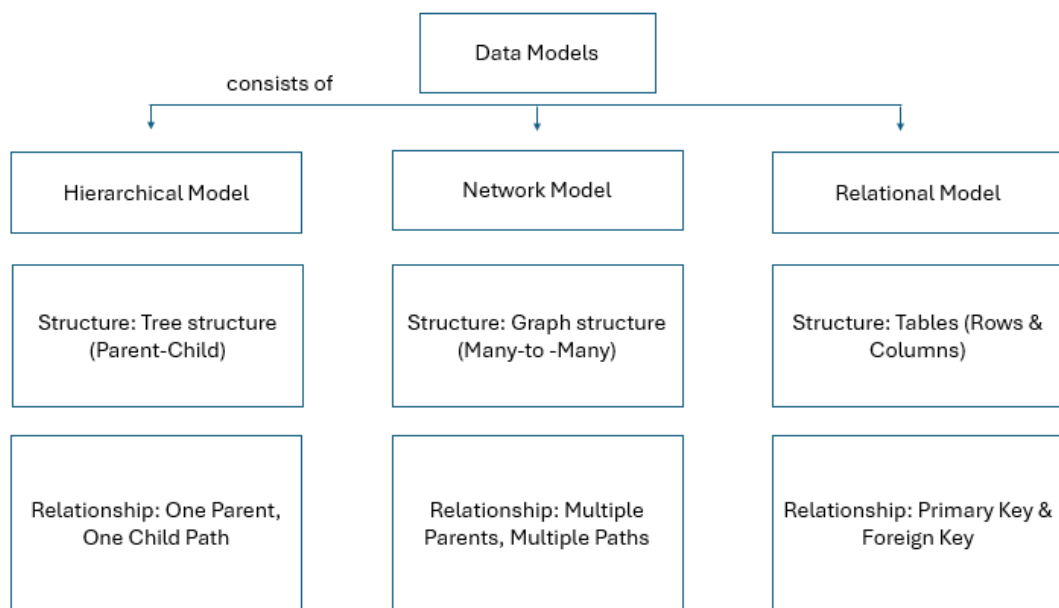


Q4. Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.



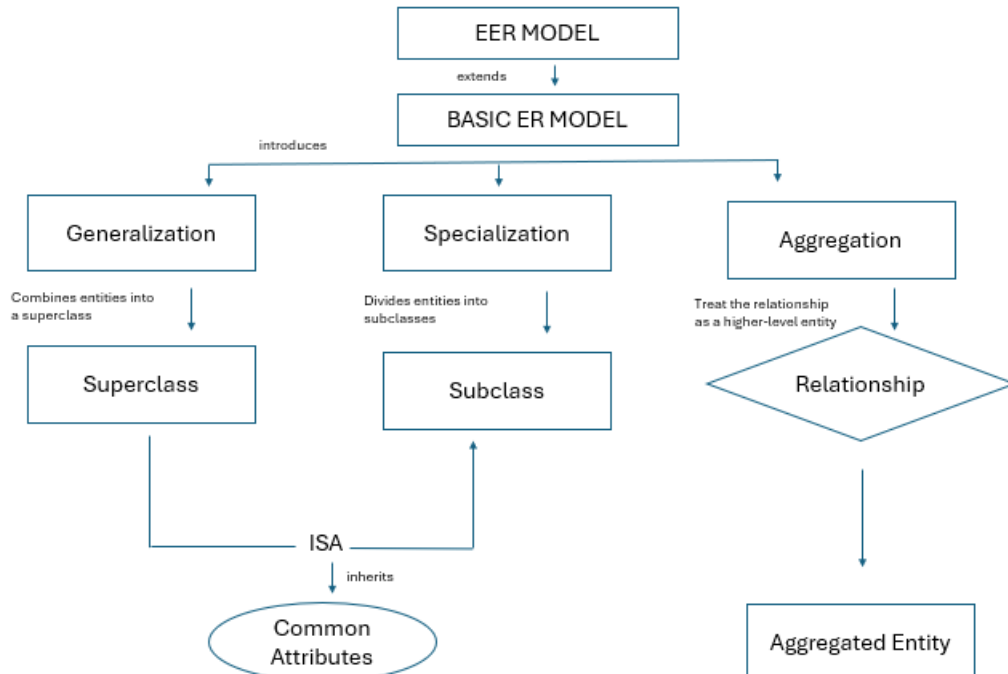


Q5. Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.

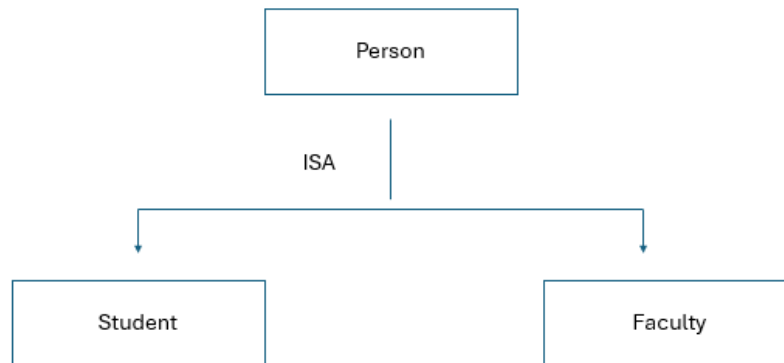


Hierarchical Model	Network Model	Relational Model
Advantages • Simple Design • Fast Access	Advantages • Flexible Links • Reduced Redundancy	Advantages • Easy SQL Queries • Data Independence
Disadvantages • Rigid Structure • No Many-to-Many	Disadvantages • Complex Design • Difficult to Manage	Disadvantages • Join Operations • Slightly Slower
Applications • File Systems • Organisation Charts	Applications • Airline Reservation • Social Networks	Applications • Banking • Hospitals

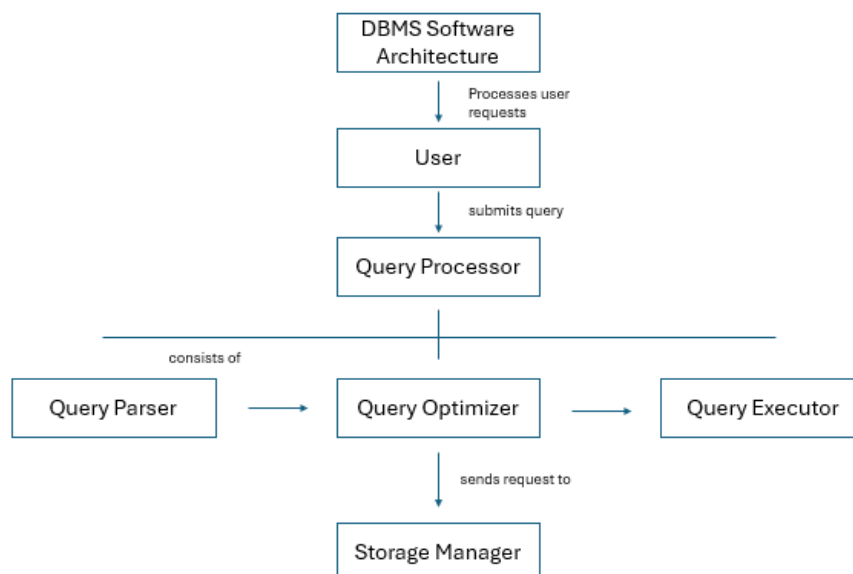
Q6. Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.

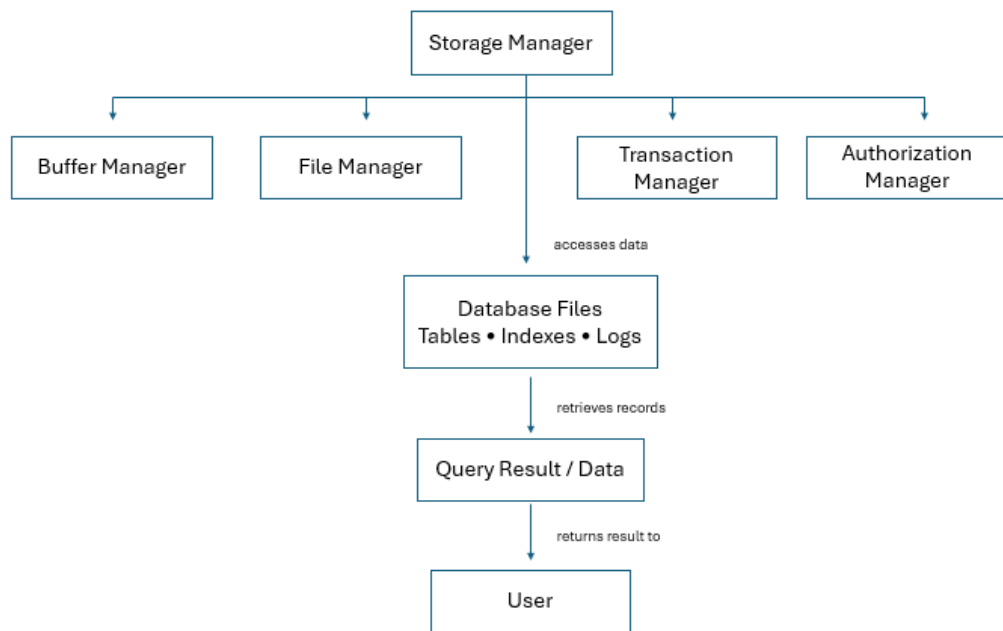


Example

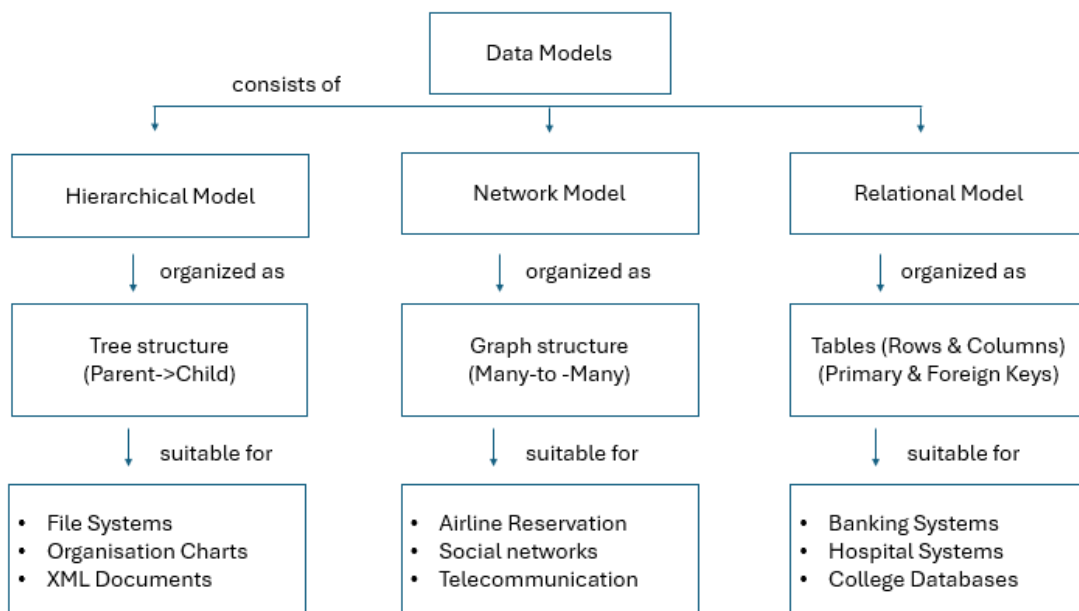


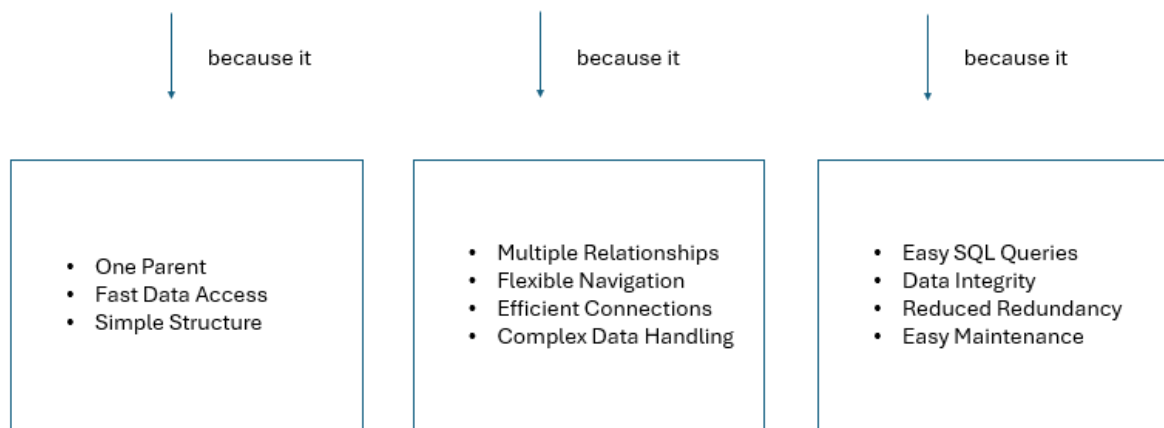
Q7. Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.



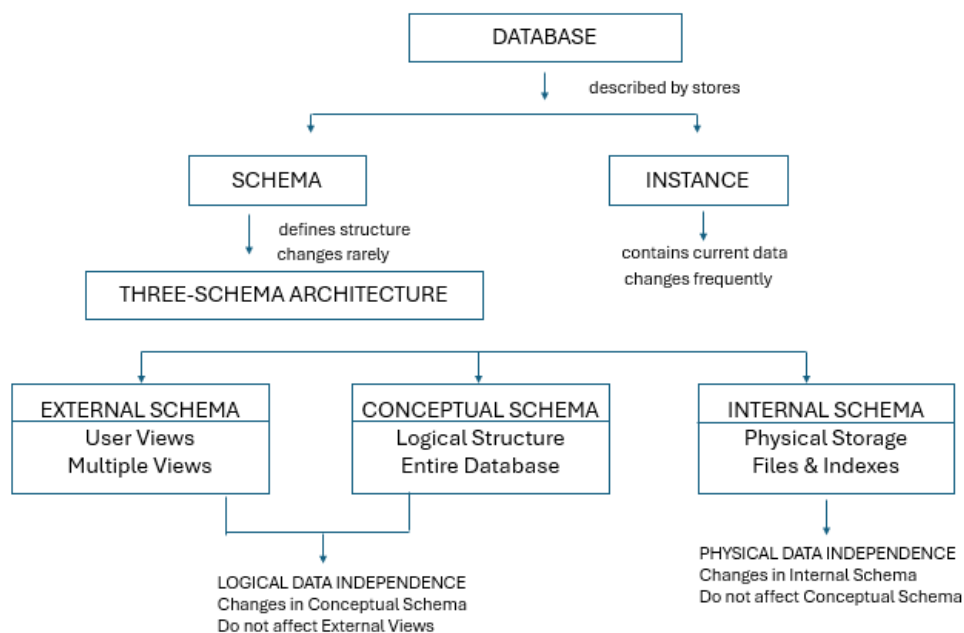


Q8. Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.





Q9. Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.



Q10. Design a concept map for the roles and responsibilities in a DBMS Environment: DBA, End User, Application Developer.

